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A data-driven optimization approach to plan smart waste collection operations

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Abstract

Waste collection is an important logistic operation that is often inefficient due to the high uncertainty associated with bin fill levels, resulting either in routes that visit empty bins or in bins overflowing due to lack of routes. To reduce such uncertainty, sensors installed in the bins can provide real-time information on waste levels. However, this is not enough, and the management of this information needs to be combined with dynamic optimization approaches to effectively design smart collection routes. This paper investigates this challenge and uses real-time information on waste levels, treated through machine learning techniques, to feed a dynamic reverse inventory routing optimization model. The resulting data-driven optimization approach allows the definition of a medium-term collection route plan that can be updated daily as new information becomes available. To demonstrate the applicability of such an approach, a real-world case is solved, and the results show significant improvements in operational efficiency and service levels.

Keywords: real-time information; dynamic reverse inventory routing problem; data-driven techniques; machine learning; hidden Markov models

1. Introduction

Emerging technologies are allowing easy access to real-time information. In this setting, transportation is being affected in different ways, as the use of information and communication technologies, when coupled with specialized routing software, can improve operations effectiveness (Golob and Regan, 2001), leading to a better use of resources (Mangiaracina et al., 2016). In this way, real-time information can be used as input for dynamic optimization approaches, enabling the design

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of a smart operation. According to Ding et al. (2020), although the definition of smart logistics is widely divergent, it usually refers to different logistics operations, such as transportation, which are planned, managed, and controlled in a more intelligent way than conventional solutions. Waste collection is an increasingly relevant issue for local administrators (Del Pia and Filippi, 2006), and an effective operation is based on the successful estimation of waste generation over time (Mitropoulos et al., 2009). In this context, if there is access to real-time information through sensorized bins, routes that focus on the bins that have higher levels of waste can be designed resulting in a reduction of unnecessary mileage traveled by the vehicles, and an increase in the cost-effectiveness of waste management systems (Mamun et al., 2016). Sensors can be used to reduce demand uncertainty, as waste accumulation depends on several factors (Nuortio et al., 2006) and is difficult to forecast. However, as collecting and transmitting the collected data usually requires a huge planning effort (Morais and Mateus, 2019), the use of remotely transmitted inventory levels of information requires a re-structuring of industry operations from fixed to dynamic routing (Ketzenberg and Metters, 2020). As stated by Thill (2000), to take advantage of real-time data, dynamic routing algorithms are needed.

The Smart Waste Collection Routing Problem (SWCRP) was first introduced by Ramos et al. (2018) to plan smart collection routes that consider real-time information on the fill levels of waste bins, obtained through volumetric sensors located inside those bins. Opposite to a static operation, where the same pre-defined routes are performed to visit all bins regardless of their fill levels, smart routes characterize a dynamic operation as new information on the level of waste is revealed daily. The smart routes are defined through a mathematical model whose objective function aims to maximize profit (given by the difference between the revenue from selling the collected recyclable waste and the transportation cost associated with the distance traveled to collect that waste) while considering the problem constraints (vehicle capacity, bin capacity, etc.). The problem was modeled and solved as a vehicle routing problem with profits (VRPP). Later, Morais et al. (2019) modeled and solved the SWCRP as a reverse inventory routing problem (IRP), where the objective function aims to maximize profit for a 10-day planning horizon (instead of maximizing profit for a single day, as happens in the VRPP). Since the IRP mathematical model uses the future fill level of the waste bins for the whole planning horizon, the authors estimate waste filling average rates from historical data using simple statistical approaches. Small-size instances were solved to compare the reverse IRP approach to the VRPP model proposed by Ramos et al. (2018), and as the main conclusion, the authors state that a medium-term planning approach is able to anticipate future events, resulting in better operational key performance indicators (KPIs - profit and amount of waste, in kg, collected per km) when compared with a short-term approach, where only the current data is used to plan routes. In this sense, the authors conclude that a short-term approach cannot look far enough ahead, resulting in bins that are visited earlier than necessary or in additional routes to attend overflowing bin. So, IRP models should be further explored and combined with predictive methods to better predict waste generation. In the waste collection literature, other works have explored the use of sensor information to plan dynamic collection routes, such as Mes (2012) and Mes et al. (2014). In these two works, however, simple statistical approaches are used to forecast demand, and exact solution methods are not explored; instead, a sequential heuristic approach is used, where first the bins to be visited are selected and then the route to visit those bins is designed.

To pursue a "smarter" operation in the waste management field and given the large amount of real-time data transmitted by sensors, efficient data storage and processing facilities are needed

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(Hashem et al., 2016). If efficiently processed, this large amount of data can be used to reduce costs and improve service levels (Ghiani et al., 2003). Instead of using statistics to forecast demand, machine learning algorithms can be used to extract patterns and knowledge from large amounts of data (Hashem et al., 2016), and a higher level of accuracy can be achieved, which is translated into better results of operational and financial performance (Feizabadi, 2020). A commonly used processing strategy is to identify a pattern in a training dataset, which is then used to classify and/or predict the behavior of new observations (testing period). To identify patterns in data, several algorithms have been developed in fields such as statistics, computer vision, and machine learning. Pattern recognition algorithms based on machine learning techniques impose few restrictions for the dataset to conform to a distribution criterion and produce patterns that are easy to understand (Bose and Mahapatra, 2001). Because not all waste bins are filled at the same rate, that is, they are characterized by a high variance, machine learning techniques can be used on individual bins to better predict their waste generation.

Given this setting, the research question to be answered is: "how to plan smarter waste collection routes using real-time information provided by sensors?" To answer this question, we extend the reverse IRP proposed by Morais et al. (2019) and develop a data-driven optimization approach (named D-DRIRP) to solve a dynamic reverse IRP. We combine mathematical routing models, real-time data from sensors, and machine learning algorithms to create a novel approach for waste collection optimization. In the D-DRIRP approach, daily data updates enable the design of a dynamic waste collection route plan that simultaneously prevent the unnecessary collection of empty bins and the occurrence of overflowing waste bins. A rolling horizon approach is used to daily update a medium-term route plan designed at the beginning of the planning horizon, according to the availability of new sensor information revealed over the planning horizon. To extract more accurate estimates from the enormous amount of available data, machine learning techniques are used. Since the amount of waste inside a waste bin (weight) is hidden and there is only access to observations (sensor readings) based on volumetric sensor measurements that have some noise associated, in this work we use hidden Markov models (HMMs) to estimate hidden variables based on observations dependent on those hidden variables.

The proposed approach is tested using an industrial case study from a Portuguese municipality that collects municipal solid waste through a "blind operation." In this type of operation, there is no access to information about bin fill levels and the waste collectors follow pre-defined static routes (i.e., the same route is repeated every given time interval) to visit all bins, regardless of their fill levels. This results in a low efficiency of the operation, as empty bins are usually visited, and resources are used for no purpose. This is where the proposed smart waste collection approach can add value to the operation, as by accessing real data on bin fill levels, it allows the planning of more efficient routes where no empty bins are visited. Recently, the company has been equipping its waste bins with volumetric sensors that transmit real-time information about the fill levels of the bins. The company aims to use this information to improve the efficiency of its collection operations and prevent overflows. We use this real-time information in the proposed D-DRIRP to design a dynamic route plan that takes into consideration the trade-off efficiency (measured by kg per km ratio) versus service level (measured by the number of overflowing bins). To compare methods to estimate future waste filling rate in this work, we consider model averaging and HMMs. The dynamic route plan designed through the application of the proposed approach is compared to the static routes currently performed by the company.

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The main contributions of this work are:

- First, we use machine learning techniques to predict the probability of a waste bin overflowing and estimate waste filling rates, improving service levels. Such techniques allow us to handle noisy data transmitted by sensors that are installed in a real-life environment.
- Second, we develop a novel data-driven approach (D-DRIRP) to plan a smarter waste collection operation by combining real-time information provided by sensors, machine learning techniques, and an exact optimization approach.
- Finally, we apply the proposed approach in a real-world case where sensor data is available, demonstrating its applicability and potential.

The remainder of this paper is structured as follows: Section 2 presents a literature review on IRP approaches and machine learning methods; next, Section 3 explains the proposed solution approach; Section 4 explores its application in a real-case study; finally, Section 5 presents the conclusions and future work.

2. Literature review

Given the characteristics of the introduced problem, this section presents a review of the related IRP and machine learning literature. Section 2.1 analyzes IRP problems and how they have been coupled with real-time data. Section 2.2 discusses the importance of adequate processing of available real-time data and how machine learning techniques have been used to predict waste generation. Finally, Section 2.3 identifies the literature gaps, states the conclusions, and highlights the contributions of this paper.

2.1. Related IRP literature

The IRP was introduced by Bell et al. (1983) to handle a real case of distribution of air products integrating decisions on when to visit a customer and how much to deliver, avoiding stockouts. Later, Dror et al. (1985) studied the effects of short-term decisions on long-term planning, using the probabilities of a customer running out on a specific day in the planning period, the average cost to deliver to the customer, and the anticipated cost of a stockout; in this work, the authors presented a single period IRP to solve the problem of propane distribution where demand is different in each period. Based on the idea of adapting previous vehicle routing problem (VRP) heuristics, Dror and Levy (1986) proposed an algorithm for a weekly IRP, applying a heuristic to reduce costs in the planning period and generating an initial solution to a VRP by keeping track of vehicle capacities and customer inventories. After this, several papers defined several variants of the problem and, according to Coelho et al. (2014), despite the large number of contributions, the integration of distribution and inventory problems proved difficult to handle. Lately, and given the increasing environmental pressures, IRP works have been focusing on reducing fuel consumption and CO_2 emissions (for works which deal with the green IRP, see e.g., Cheng et al., 2017; Soysal et al., 2018; Alinaghian et al., 2021).

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By integrating inventory management with routing problems over a given time horizon, the IRP addresses three different decisions: when to visit a client, how much to deliver to or collect from this client, and how to combine clients in a route (Baita et al., 1998; Coelho et al., 2014). In this problem, the trade-off between inventory cost and transportation cost must be considered by the decision-maker to optimize vehicle routes to deliver products (in the case of a forward IRP) or to collect products (in the case of a reverse IRP). The classical IRP aims to minimize the sum of the inventory cost and the distribution/collection cost to determine which customers should be visited and the amount to supply to or collect from each selected customer (the inventory allocation decision). For this inventory allocation purpose, routing cost information is needed so that the marginal delivery/collection cost for each customer can be accurately computed. On the other hand, routing decisions directly influence inventory holding costs. The delivery/collection cost for each customer depends on the vehicle routes, which in turn requires information about customer selection and the amount of inventory allocated to/collected from each customer (Moin and Salhi, 2007).

A distinction must be made between a forward IRP and a reverse IRP. In the forward IRP, the goal is to optimize the vehicle routes to deliver products as the customer inventory level decreases over time, while in the reverse IRP, the route optimization is made considering the collection of products as the customer inventory level increases over time. In addition to this, and according to Le Blanc et al. (2004), in the forward IRP, the increased efficiency comes from a well-balanced trade-off between inventory and transportation costs, while in the reverse IRP, the expected efficiency gains come from foreseeing when a collection should take place and from actively searching for possible combinations in planning collection trips, reducing the overall transportation cost (this is because the reverse IRP generally presents a low inventory holding cost and a high transportation cost, compared to the product value, contrary to the forward IRP, which is characterized by a high inventory holding cost and a moderate to low transportation cost, compared to the product value). The IRP has recently gained attention, but while the forward IRP has been much explored (see reviews written by Baita et al., 1998; Moin and Salhi, 2007; Andersson et al., 2010; Coelho et al., 2014), the reverse IRP has been barely investigated (see Aksen et al., 2012; Mes et al., 2014; Nolz et al., 2014; Elbek and Wøhlk; 2016; Morais et al., 2019).

When considering demand information, Coelho et al. (2014) classify the IRPs into three distinct groups: deterministic IRPs, stochastic IRPs (SIRPs), and dynamic IRPs (DIRPs). Deterministic IRPs assume that demand information is known by the decision-maker at the beginning of the planning horizon; SIRPs consider a demand probability distribution; and DIRPs assume that demand is not completely known at the beginning of the planning horizon but will be gradually revealed over time. Deterministic IRPs use known constant demand usage rates (Campbell et al., 1997), characterizing a static strategy. In some cases, demand can be defined by a stable rate and suppliers know those customers' demand rates. In the case of SIRPs, demand is assumed to follow some probabilistic distribution. Due to demand uncertainty, penalties are usually imposed to avoid any shortages/overflows that may occur. Essentially, the goal of the SIRP is the same as in the deterministic case but is defined to accommodate stochastic and unknown future parameters. Due to the size, complexity, and stochastic nature of the problem, SIRP solutions are usually computationally hard to find. When considering the DIRP, the literature on this field deals mainly with IRPs where demand varies across the time horizon. The solution to these IRPs consists of developing a dynamic approach that considers route planning at the beginning of several periods. For each demand variation, the IRP solution is re-optimized, or a new IRP is solved.

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When considering DIRP models coupled with real-time information, several difficulties arise since the dynamism required to deal with information that is updated every day is not straightforward. However, nowadays DIRP approaches are necessary as means to explore the access to real-time information and to better support the decision-making processes of operations. Such works are not easily found in the literature applied to waste collection. So far, and to the best of our knowledge, only three works have been published exploring this problem: Mes (2012), Mes et al. (2014), and Morais et al. (2019). Mes (2012) uses discrete event simulation to evaluate how dynamic planning methodologies can be applied to collect waste from underground containers equipped with motion sensors, which count the number of lid openings, recording how many disposals are made. The author compares a static planning methodology based on a desired time between emptying, with a dynamic planning, where containers are daily selected based on their estimated fill levels. For dynamic planning, a heuristic is used, and containers are divided into MustGos (those that need to be emptied on the current day) and MayGos (those which are only included in routes if there is space left in truck capacity and working hours). The MustGos are first included using the cheapest insertion heuristic, and then MayGos would be inserted if space in the trucks is available. A real case is solved and the author concludes that major improvements are achieved when assuming a dynamic waste collection policy. Mes et al. (2014) consider a reverse IRP for collecting waste from underground containers where sensors are installed to transmit fill-level information. In their case, the company must plan when to empty the containers and how to route the vehicles, such that the collection costs are minimized. To solve a two-week SIRP for waste collection dynamically, the authors develop a parameterized heuristic considering the long-term impact by using adjustable parameters and historical cost information. The tunable parameters determine, on a given day, the containers that are eligible for emptying and the cheapest insertion procedure is used to form routes to visit the eligible containers. The authors tune the parameters via simulation. The heuristic approach is then applied to an operational planning problem where the goal is to decide which containers to empty. Morais et al. (2019) develop a mixed integer linear programming model for the recyclable waste collection IRP (reverse IRP), where the objective function aims to maximize profit for a medium-term planning horizon. The authors simulate the use of real-time information about the fill levels of the bins, allowing the definition of smart waste collection routes. However, they do not explore in their work any predictive methods for waste generation estimation or the actual use of real-time sensor data.

2.2. Related machine learning literature

The Internet of Things (IoT) enables the access to vast volumes of data. When placing sensors in any kind of "thing," like a waste bin, a large amount of real-time data becomes available to the decision-maker. Consequently, because such enormous volumes of data become the core of the services rendered by the IoT, it is important to obtain valuable insights from this large amount of data (Hashem et al., 2016). In this context, machine learning techniques are finding wide popularity for data analysis and pattern recognition (Bose and Mahapatra, 2001). By replacing time-consuming human activity with automatic techniques that improve accuracy, machine learning aims to provide increasing levels of automation in the knowledge process by discovering and exploiting regularities in training data.

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According to Langley and Simon (1995), machine learning is the study of computational methods to automate the process of knowledge acquisition from experience, where most evaluation is experimental in nature, aimed at showing that the learning method leads to performance on a separate test set that is better than performance on that test set without learning. According to Davenport and Ronanki (2019), cognitive insights provided by machine learning differ from those available from traditional analytics in three ways: they are much more data-intensive and detailed, the models typically are trained on some part of the dataset, and the models get better—that is, their ability to use new data to make predictions or put things into categories improves over time. Because of this, machine learning applications are being used extensively in many important daily activities, such as to predict what a particular consumer is likely to buy; to identify credit fraud in real time and detect insurance claims fraud; to analyze warranty data to intensify safety or quality problems in automobiles and other manufactured products; to automate personalized targeting of digital ads; and to provide insurers with more-accurate and detailed actuarial modeling (Davenport and Ronanki, 2019).

Fueled by the growing focus on municipal solid waste and supported by methodological innovations, a growing number of studies aim at assessing the efficiency of local municipal solid waste collection services (Rogge and De Jaeger, 2013). In this sense, the efficiency of waste collection operations can be improved if waste filling rates can be accurately estimated. Therefore, estimates about the amount of waste to be deposited in each waste bin are necessary and should be acquired from the enormous amount of available historical data using proper data analysis and pattern recognition techniques. In the literature, such predictions have been addressed by several authors using different machine learning techniques (see the works of Chen and Chang, 2000; Tesfahun et al., 2016; Johnson et al., 2017). To predict waste generation from a sample characterized by a high variance, as is the case of waste filling rates for individual bins, standard dynamical modeling methods, namely, HMMs or dynamic linear models, such as auto regressive models with an exogenous vector process (ARX) and auto regressive moving average (ARMA, Kumar and Jain, 2010), are commonly used. While the first method provides a more robust statistical analysis of the data, the second one can provide a more direct estimation of the continuous quantities.

To predict waste generation in megacities that strive to achieve sustainable development, Jiang and Liu (2016) propose a model-driven statistical learning approach that dynamically traces fluctuations caused by uncertainties associated with waste generation. This approach combines three techniques: wavelet denoising, Gaussian mixture models, and HMMs, to forecast the dynamic generation of municipal waste under uncertainties. In their work, the forward algorithm (Rabiner and Juang, 1986) is used to search the historical data and find the data pattern most similar to the current data, in order to forecast the future trend of the periodic fluctuation problem. The proposed approach is tested in the case studies of Shanghai and Beijing's annual municipal solid waste generation, and the results indicate that the HMM approach generally outperforms approaches like ARMA and auto regressive integrated moving average (ARIMA), among others, for solving the municipal waste generation forecasting problem. In our study, we chose HMMs to estimate waste generation, as this approach appears in the literature as the most adequate for the waste generation forecasting problem and allows the creation of a decision rule that establishes a compromise between service level and efficiency, two performance indicators that are explored throughout this study.

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2.3. Literature gaps and contributions

From the literature review, it is possible to conclude that most of the IRP literature deals with distribution problems, while collection problems have been barely explored, especially when considering waste collection. According to Fleischmann et al. (2000), the major distinguishing factor between product recovery and traditional distribution networks is, in a wide sense, supply uncertainty (which can be reduced using modern information technologies such as sensors).

In this context, only a few works are identified that explore DIRP approaches applied to waste collection where real data transmitted by installed sensors are used (and sensor data are not simulated). Table 1 highlights the differences between this and previous work. Using the structural classification scheme proposed by Coelho et al. (2014), we observe that the problem studied in most of the identified works had similar characteristics: finite time horizon, one-to-many structure (vehicles travel from one depot to visit several bins), multiple routing (several bins are visited in the same route), order-up-to level inventory policy (whenever a bin is visited, it is fully emptied), and nonnegative inventory decisions (bins are not allowed to overflow and must be visited before it happens). However, from all the identified works, only Mes (2012) and Mes et al. (2014) use real-sensor data to design smart waste collection routes. In the published works, cost minimization is typically assumed, and only Morais et al. (2019) attempt to maximize the profit associated with the waste collected. This work is also the only identified work that developed an integrated optimization approach in which the selection of bins to collect and the routing decisions are addressed simultaneously. The other analyzed works explore a sequential approach (select first-route second) where heuristics or metaheuristics are combined to solve the DIRP. To explore the existing gaps, the present work uses real data transmitted by sensors installed in an industrial environment and studies a data-driven approach that embeds a prediction method able to estimate different accumulation rates depending on the amount of waste inside a bin when the sensor transmission occurs. In the present work, the exact optimization approach proposed by Morais et al. (2019) is extended to develop a data-driven approach that defines smarter waste collection routes using prediction based on real-time sensor information.

3. A data-driven approach to solve a dynamic reverse IRP

The D-DRIRP approach combines predictive methodologies (to predict waste generation) and optimization methods to define a dynamic collection route plan. This is illustrated in Fig. 1, where it can be seen how real-time sensor information about the fill levels of waste bins feeds the D-DRIRP approach. Sensor information is also used as historical data to feed predictive models (presented in Section 3.2). These models estimate an expected daily filling rate, required by the reverse IRP model to predict the fill levels of waste bins within the planning horizon. The dynamic reverse IRP (presented in Section 3.1) is composed of a dynamic rolling horizon approach that is used to solve the reverse IRP mathematical model.

In this section, first, the dynamic reverse IRP is detailed, and then the predictive models are explored. At the end of this section, the integration of the predictive methods into the dynamic reverse IRP is explained.

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Table 1
Comparison between dynamic inventory routing problem (DIRP) approaches applied to waste collection

	DIRP approaches for waste collection									
	Time horizon	Structure	Routing	Inventory policy	Inventory decisions	Sensor info	Demand forecast	Objective function	Operational decisions	Solution method
Mes (2012)	Finite	One-to-many	Multiple	Order-up-to	Back-ordering	x	Estimated mean	Min costs	Sequential	Heuristics
Mes et al. (2014)	Finite	One-to-many	Multiple	Order-up-to	Back-ordering	x	Estimated mean	Min costs	Sequential	Heuristics
Nolz et al. (2014)	Finite (rolling)	One-to-many	Multiple	Order-up-to	Nonnegative		Estimated mean	Min costs	Sequential	Heuristics
Morais et al. (2019)	Finite (rolling)	One-to-many	Multiple	Order-up-to	Nonnegative		Estimated mean	Max profit	Integrated	Exact
Akkerman et al. (2020)	Finite (rolling)	One-to-many	Multiple	Order-up-to	Back-ordering		Estimated mean	Min costs	Sequential	Approximate
This work	**Finite (rolling)**	**One-to-many**	**Multiple**	**Order-up-to**	**Nonnegative**	**x**	**Overflows prediction**	**Max–min weight/distance**	**Integrated**	**Exact**

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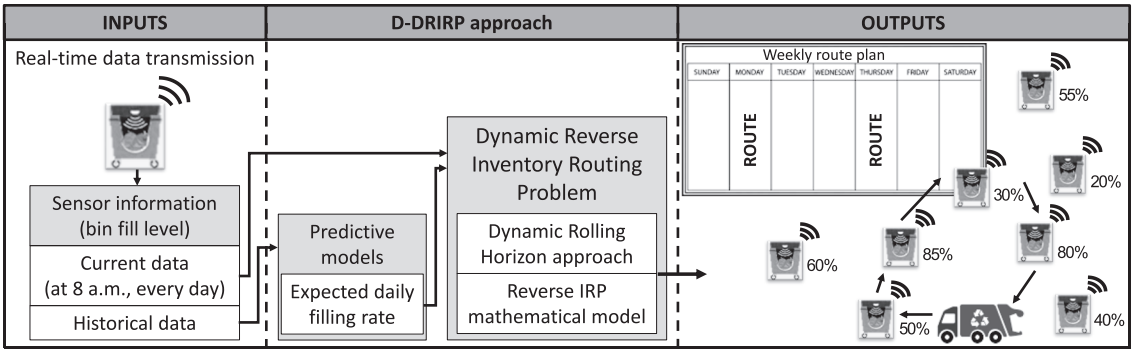

Fig. 1. A data driven dynamic reverse inventory routing problem (D-DRIRP) approach.

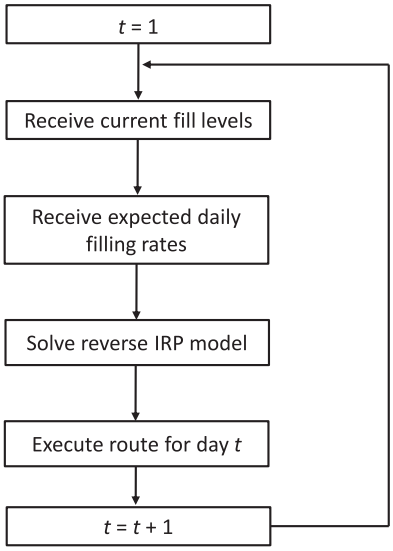

Fig. 2. Dynamic reverse inventory routing problem (IRP).

3.1. Dynamic reverse IRP

The dynamic reverse IRP embeds a reverse IRP mathematical model that aims to maximize the amount of waste collected while minimizing the transportation distance. On the morning of day t, after receiving the sensor information about the current fill levels of the bins and the expected daily filling rate for the days ahead (the latter given by the predictive models as explained in Section 3.2), the reverse IRP mathematical model is solved and an initial optimal routing plan for a given planning horizon is defined. This plan defines which days and which routes to operate, which waste bins to visit, and which sequence to follow. On the next iteration, that is, the next day ($t = t + 1$), as new real-time information becomes available, the initial medium-term routing plan is updated through a dynamic rolling horizon approach, where the reverse IRP mathematical model is again solved. This is schematically shown in Fig. 2.

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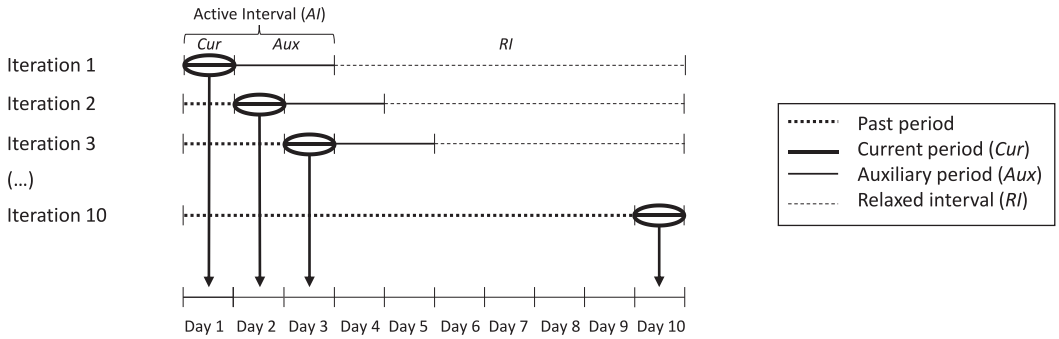

Fig. 3. Rolling horizon approach exemplified for a 10-day planning period, where $AI = 3$ days and $RI = 7$ days.

In this work, new sensor information is available at most once a day, so these new data are used every day to update a medium-term route plan. In this sense, no replanning is done once the routes start to be performed. Different than a continuous data updating, where routes are changed in an ongoing fashion and technological support is necessary to permit real-time communication between the vehicles and the decision-maker, we explore daily data updating to keep the solution closer to the practice of the company considered in this work. Their practice is to define the routes at the beginning of each day and then execute them.

When solving the reverse IRP mathematical model in a rolling horizon fashion, the planning period is divided into two intervals: an active interval (*AI*), which includes the days considered for the original binary variables (that are about to be presented next), and a relaxed interval (*RI*), which includes the remaining days and where the binary variables are relaxed, assuming positive values. As can be seen in Fig. 3, the active interval consists of a current period (*Cur*), which corresponds to the first day of the planning horizon (i.e., being solved at the moment), day t, and an auxiliary period (*Aux*), which helps the model to consider future events. The auxiliary period length is variable and, as stated by Meira et al. (2018), its function is to guide the model, helping to anticipate future problems. The resulting plan is executed for day t, and the next iteration is carried out on day $t + 1$.

Using the rolling horizon approach, the mathematical model is solved for each day within the planning horizon. Each time the mathematical model is solved, different routes are planned for the current day and the following days. The route planned for the current period (current day) is performed, and after receiving new information from the sensors, on the following days, the remaining routes are replanned. Planning the routes for the auxiliary period and relaxed period (although these routes are updated on the following days) allow the model to see further ahead in the time horizon and make better decisions in the current period.

The mathematical formulation of the reverse IRP model is presented below. This problem can be defined as follows: given a set of n waste bins and a depot (where the vehicle starts and ends the routes), a complete undirected graph is defined on $n + 1$ nodes, with distances d_{ij} for each edge (i,j) in the graph. Each waste bin i has a maximum capacity E_i (in m^3) and the vehicle has a fixed capacity Q (in kg). At the beginning of each day t, the sensors located inside the waste bins transmit information about the fill levels of the bins, in the percentage of volume. This information is translated into kilogram, given the waste density (B) and the waste bin capacity (E_i). A parameter

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S_{i1} that represents the fill-level information transmitted by the sensor located inside waste bin I on the first day ($t = 1$) of the planning horizon (e.g., at 8:00 a.m.) is defined. The expected daily filling rate $\hat{a}_{it}$ for each day $t \in T$ is computed through predictive methods based on historical data transmitted by the sensors (see Section 3.2).

Contrary to Morais et al. (2019), who considered profit maximization (as their case study regards recyclable waste, which has market value and, consequently, associated revenues), in this paper, we work with municipal/undifferentiated waste, which has no market value. As the aim of the company is to provide an effective waste collection service, the goal in this paper is to maximize the amount of waste collected (kilograms) while minimizing the transportation distance to be traveled (kilometers). To handle the fact that both objective terms have different units, we explore scalarization techniques. We assume a 50–50 weight based on consultations with the operations manager of the company involved in this research, who claim that the amount of waste collected has the same weight for the company as the cost incurred on the defined routes. To apply the scalarization technique, we had to compute the minimum and maximum values for both objectives (weight and distance). The minimum values are both zero, while the maximum values are given by parameters C and R. Parameter C represents the sum of the total weight collected if all waste bins are collected at full capacity every day within the planning horizon. Parameter R represents the total distance traveled if a route where all waste bins are visited is followed every day within the planning horizon.

Index sets

$I = \{1, 2, \ldots, n{+}1\}$	Set of n waste bins and the depot $n + 1$
$T = \{1, 2, \ldots, h\}$	Set of h days in the planning horizon
$M = \{1, 2\}$	Set of moments of the day: 1—beginning of the day, 2—end of the day

Parameters

Q	Vehicle capacity (in kg)
B	Waste density (in kg/m^3)
d_{ij}	Distance between node i and node j (in km)
S_{i1}	Fill level of bin i at the beginning of the first day ($t = 1$) of the planning horizon (which is the information given by the sensor [in the percentage of volume])
$\hat{a}_{it}$	Expected daily filling rate of bin i for each day t (in percentage of volume)
E_i	Capacity of bin i (in m^3)
ε	Very small nondimensional value
AI	Active interval for the rolling horizon (in number of days)
RI	relaxed interval for the rolling horizon (in number of days)
C	Total weight collected if all waste bins would be collected at full capacity every day within the planning horizon (in kg)
R	Total distance traveled if a route where all waste bins are visited would be followed every day within the planning horizon (in km)

Decision variables

x_{ijt}	Binary variable indicating if edge (i,j) is visited on day t, $(i,j \in \mathrm{I})$ $(t \in \mathrm{T} \setminus \{t > AI\})$
x'_{ijt}	Positive variable indicating if edge (i,j) is visited on day t, $(i,j \in \mathrm{I})$ $(t \in \mathrm{T} \setminus \{t \leq AI\})$
y_{it}	Binary variable indicating if waste bin i is visited on day t, $(i \in \mathrm{I})$ $(t \in \mathrm{T} \setminus \{t > AI\})$

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y'_{it} Positive variable indicating if waste bin i is visited on day t, ($i \in$ I) ($t \in$ T\ $\{t \leq AI\}$)
f_{ijt} Positive variable representing the flow between node i and node j on day t, ($i,j \in$ I) ($t \in$ T)
w_{it} Positive variable representing the amount of waste collected at waste bin i on day t, ($i \in$ I) ($t \in$ T)
u_{itm} Positive variable representing the amount of waste in bin i at moment m on day t, ($i \in$ I) ($t \in$ T) ($m \in$ M)

Model formulation:

Objective function

$$\max Z = \left(\left(\sum_{i \in I \setminus \{n+1\}} \sum_{t \in T} w_{it}\right) \Big/ C - \left(\left(\sum_{i \in I} \sum_{j \in I} \sum_{t \in T} x_{ijt} d_{ij} + \sum_{i \in I} \sum_{j \in I} \sum_{t \in T} x'_{ijt} d_{ij}\right) \Big/ R\right)\right); \quad (1)$$

subject to the following constraints:

$$\sum_{j \in I, (j \neq i)} f_{ijt} - \sum_{j \in I, (j \neq i)} f_{jit} = w_{it}, \ \forall i \in I \setminus \{n+1\}, \ t \in T \quad (2)$$

$$f_{ijt} \leq \left(Q - \hat{a}_{jt}\right)\left(x_{ijt'} + x'_{ijt''}\right), \ \forall i, j \in I, t \in T, t' \in T \setminus \{t > AI\}, t'' \in T \setminus \{t \leq AI\}, i \neq j; \quad (3)$$

$$f_{ijt} \leq Q - w_{jt}, \ \forall i \in I, j \in I \setminus \{n+1\}, t \in T, i \neq j; \quad (4)$$

$$f_{ijt} \geq w_{it} - BigM\left(1 - x_{ijt}\right), \forall i \in I \setminus \{n+1\}, j \in I, t \in T \setminus \{t > AI\}, i \neq j; \quad (5a)$$

$$f_{ijt} \geq w_{it} - BigM\left(1 - x'_{ijt}\right), \ \forall i \in I \setminus \{n+1\}, j \in I, t \in T \setminus \{t \leq AI\}, i \neq j; \quad (5b)$$

$$\sum_{j \in I, (j \neq i)} x_{jit} + \sum_{j \in I, (j \neq i)} x'_{jit'} = y_{it} + y'_{it'}, \ \forall i \in I \setminus \{n+1\}, t \in T \setminus \{t > AI\}, t' \in T \setminus \{t \leq AI\}; \quad (6)$$

$$\sum_{j \in I, (j \neq i)} x_{ijt} + \sum_{j \in I, (j \neq i)} x'_{ijt'} = y_{it} + y'_{it'}, \ \forall i \in I \setminus \{n+1\}, t \in T \setminus \{t > AI\}, t' \in T \setminus \{t \leq AI\} \quad (7)$$

$$\sum_{i \in I \setminus \{n+1\}} x_{ijt} + \sum_{i \in I \setminus \{n+1\}} x'_{ijt'} = \sum_{i \in I \setminus \{n+1\}} x_{jit} + \sum_{i \in I \setminus \{n+1\}} x'_{jit'}, \ \forall j \in I : \{j = n+1\},$$
$$t \in T \setminus \{t > AI\}, t' \in T \setminus \{t \leq AI\} \quad (8)$$

$$w_{it} \leq (E_i B)\left(y_{it'} + y'_{it''}\right), \ \forall i \in I \setminus \{n+1\}, t \in T, t' \in T \setminus \{t > AI\}, t'' \in T \setminus \{t \leq AI\}; \quad (9)$$

$$u_{itm} \leq BigM\,(1 - y_{it}) + \hat{a}_{it}, \forall i \in I \setminus \{n+1\}, t \in T \setminus \{t > AI\}, m \in M : \{m = 2\}; \quad (10a)$$

$$u_{itm} \leq BigM\left(1 - y'_{it}\right) + \hat{a}_{it}, \ \forall i \in I \setminus \{n+1\}, t \in T \setminus \{t \leq AI\}, m \in M : \{m = 2\}; \quad (10b)$$

$$u_{itm} = S_{it'} + u_{it''-1m'}, \ \forall i \in I \setminus \{n+1\}, t \in T, t' \in T : \left\{t' = 1\right\}, t'' \in T : \{t'' > 1\},$$
$$m \in M : \{m = 1\}, m' \in M : \left\{\mathrm{m}' = 2\right\}; \quad (11)$$

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$$u_{itm} = u_{itm'} + \hat{a}_{it} - w_{it},\ \forall i \in I \backslash \{n+1\}, t \in T, m \in M : \{m = 2\}, m' \in M : \left\{m' = 1\right\}; \quad (12)$$

$$u_{i'tm} = \sum_{i \in I \backslash \{n+1\}} w_{it},\ \forall i' \in I : \left\{i' = n+1\right\},\ t \in T,\ m \in M : \{m = 2\} \quad (13)$$

$$w_{it} \leq u_{itm},\ \forall i \in I \backslash \{n+1\}, t \in T, m \in M : \{m = 1\}; \quad (14)$$

$$u_{itm} \leq E_i B,\ \forall i \in I \backslash \{n+1\}, t \in T, m \in M : \{m = 2\} \quad (15)$$

$$y_{it'} + y'_{it''} \geq \left(\frac{(u_{itm} - E_i B)}{E_i B}\right) + \varepsilon,\ \forall i \in I \backslash \{n+1\}, t \in T, t' \in T \backslash \{t > AI\},$$
$$t'' \in T \backslash \{t \leq AI\}, m \in M : \{m = 1\}; \quad (16)$$

$$x_{ijt}, y_{it} \in \{0, 1\},\ \forall i, j \in I, t \in T, i \neq j; \quad (17)$$

$$x'_{ijt}, y'_{it} \in [0 - 1],\ \forall i, j \in I, t \in T, i \neq j; \quad (18)$$

$$f_{ijt},\ w_{it},\ u_{itm} \in \mathrm{R}^+,\ \forall i, j \in I, t \in T, \mathrm{m} \in M, i \neq j. \quad (19)$$

The objective function (1) considers the maximization of the difference between the amount of collected waste and the total distance traveled to collect that waste. As both terms represent different goals, this equation is normalized to assure that the coefficients of the model are scaled with respect to the units of the inputs. Both terms are divided by the maximum values of collected waste and distance traveled.

Flow balance is given by constraints (2), (3), (4), (5a), and (5b):

- Constraints (2) represent the flow balance at each waste bin i.
- Constraints (3) and (4) provide upper bounds on the flow variables f_{ijt} by considering the vehicle capacity and the amount of waste collected from node j when a vehicle travels from i to j on day t.
- Constraints (5a) and (5b) provide lower bounds on the flow variables by ensuring that if a vehicle travels from i to j on day t, all the waste accumulated in bin i must be collected.

Degree balance is guaranteed by constraints (6), (7), and (8):

- Constraints (6) and (7) link variables x_{ijt} and y_{it} and give the incoming and outgoing degree balance for each waste bin i, ensuring that the incoming/outgoing degree of waste bin i must be equal to 1, if waste bin i is visited on day t and equal to 0 otherwise.
- The depot's degree balance constraints (8) set that the incoming and outgoing degrees must be the same.

Inventory is calculated using constraints (9), (10a), (10b), (11), (12), (13), (14), (15), and (16):

- Constraints (9) ensure that the collection amount at waste bin i on day t must be zero unless it is visited on that day.

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- Constraints (10a) and (10b) guarantee that the inventory at waste bin i must be equal to the expected daily filling rate at the end of day t, if it is visited on that day.
- Constraints (11) ensure that the initial inventory at waste bin i on day 1 is equal to the information sent by the sensors about the fill level of the waste bin. On the subsequent days, the initial inventory at waste bin i is equal to the inventory at waste bin i at the end of the previous day.
- Constraints (12) update the final inventory at waste bin i on day t by incorporating the expected daily filling rate of bin i on day t, namely, $\hat{a}_{it}$, and the amount of waste collected at waste bin i on day t, w_{it}. If waste bin i is not visited on day t, then w_{it} will be zero and the initial inventory increases by $\hat{a}_{it}$.
- Inventory balance constraints for the depot (13) consider the total collected waste amount, and constraints (14) force the whole initial inventory at bin i on day t to be collected when this waste bin is visited.
- Constraints (15) guarantee that the final inventory at bin i is not higher than the bin capacity.
- Constraints (16) ensure that when the amount of waste in bin i at the beginning of day t is equal to or higher than the waste bin capacity, this waste bin must be visited, so variables y_{it} and y'_{it} assume values greater than zero,and that when the amount of waste in bin i at the beginning of day t is lower than the waste bin capacity, this waste bin can be visited or not.

Finally, constraints (17) to (19) give the domain of the variables.

3.2. Predictive methodologies

For the application of predictive methods, historical observations (training period) are used to predict the behavior of new observations (testing period). To assess the future fill levels of the waste bins, the developed reverse IRP model incorporates estimates about the daily waste filling rates for the whole testing period (parameter $\hat{a}_{it}$, $\forall i \in I\backslash\{n+1\}$, $t \in T_{testing}$). To compute these estimates, a training period $T_{training}$ is considered for the application of predictive methods that process historical sensor readings about the fill levels of the waste bins (S_{it}, $\forall i \in I\backslash\{n+1\}$, $t \in T_{training}$). Two different data-driven predictive methodologies are considered: (1) a daily average method and (2) HMMs.

3.2.1. Daily average method

The first predictive method consists of simply computing a daily average filling rate for each waste bin $i \in I\backslash\{n+1\}$ and for each day $t \in T_{training}$. To this end, the volume difference between sensor readings of two consecutive days is computed and considered. We assume that the bins are completely emptied during collection and are then filled at a rate that is reflected in the information transmitted by the sensor the next day. Since we are using historical data from sensor readings from a real-case study (where collection routes have occurred), the computation of the daily average filling rate must be carefully addressed. If no collection between those two consecutive days (t and $t+1$) has occurred, the filling rate of day t is equal to $S_{it+1} - S_{it}$. When a collection has occurred on day t, the filling rate of day t is equal to the sensor reading of the next day (S_{it+1}), as it is considered that the waste bin was completely emptied during collection and then filled up throughout the day until the next sensor reading, which occurs on day $t+1$. In this way, for each waste bin, a filling

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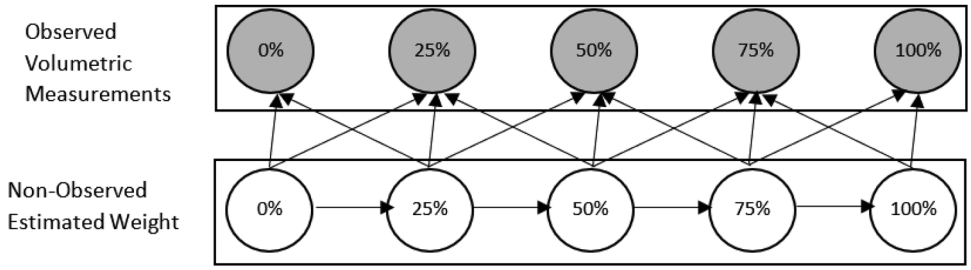

Fig. 4. Hidden Markov model representation of a waste bin.

rate for each day of the training period is computed. These daily filling rates are aggregated based on the day of the week, generating an estimated weekly day average filling rate for each waste bin (i.e., Monday's average filling rate, Tuesday's average filling rate, etc.). This takes seasonality into consideration, as this is implicitly part of the historical data.

3.2.2. HMMs

HMMs (Rabiner and Juang, 1986) are a well-established methodology for estimating hidden variables based on observations dependent on those hidden variables. In this paper, the amount of waste inside a waste bin (weight) is hidden as there is only access to noisy observations (sensor readings) based on volumetric sensor measurements (see Fig. 4). Formally, a set of states $\boldsymbol{S}$ is defined where each particular state $\boldsymbol{s}$ corresponds to a given percentage of waste present inside the waste bin. Five states are considered in the model (0%, 25%, 50%, 75%, and 100%). A state equal to 100% corresponds to an overflowing waste bin, which is a situation to avoid. The model is represented in Fig. 4, where the (real) hidden state is the amount of waste in the waste bin. The transitions between the hidden states correspond to waste deposited. It is assumed that, when waste deposition occurs, no more than 25% of the capacity of the waste bin is deposited. A transition from any of the states to the 0% state occurs when a collection is made. Information transmitted by the sensors may be inaccurate. Based on the observed data, we noticed that sensor readings present a noise level of around 25% of the actual value. So, any observation $\boldsymbol{O}$ can be perceived for the corresponding state, but it is also possible to have an observation corresponding to any of the neighboring states.

The HMM is then defined with the transition probabilities that measure the probability of one state (s_t) changing to another state (s_{t+1}) in a one-time step:

$$p(S = S_{t+1}|S = S_t). \tag{20}$$

The observation probabilities that translate the probability of making a particular observation (o_t) in a given state (s_t) is given by:

$$p(O = O_t|S = S_t). \tag{21}$$

These probabilities are estimated using historical data through the Baum–Welch algorithm (Baum and Petrie, 1966). The observation probabilities are the same for all waste bins, that is, we assume that all sensors are equal and were installed inside the waste bins in the same way. The

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transition probabilities are different for each waste bin, that is, we assume that each waste bin is filled at a specific rate.

Given the estimated model probabilities and some observations, we can compute the most probable state at a given time, where the probabilities $p(S = s_t, O = o_t, o_{t-1}, \ldots, o_{t-n})$ are obtained using the forward algorithm (the forward algorithm was proposed by Rabiner and Juang (1986) and is given by Equation (22)). In the end, the probability distribution over the states (real weight in the bins) is obtained. In the case of having a single measure, we get:

$$\begin{aligned} p(S = s_t, O = o_t) &= \sum_{s_{t-1}} p(S = s_t, O = o_t, S = s_{t-1}) \\ &= \sum_{s_{t-1}} p(O = o_t | S = s_t, S = s_{t-1})\, p(S = s_t | S = s_{t-1}) p(S = s_{t-1}), \end{aligned} \tag{22}$$

where it is possible to observe that the probability of a given state depends on the distribution of states in the previous step $p(S = s_{t-1})$, the probability of filling $p(S = s_t | S = s_{t-1})$, and the probability of observing the current measurement $p(O = o_t | S = s_t, S = s_{t-1})$.

A decision rule based on a standard probabilistic decision-making can then be established based on the cost of collecting a particular bin. If the collection cost is c and the overflow cost is v, the probability of overflow times the cost of an overflow is given by $vp(S = 100\%)$. If this value is bigger than the cost of collecting that bin, then it is better to collect it:

$$\begin{aligned} &vp(S = 100\%) \geq c \\ &p(S = 100\%) \geq c/v \end{aligned} \tag{23}$$

For example, given a 90% probability of a waste bin being full on the following day, the best decision is to collect it if the ratio of collection cost to overflow cost is less than or equal to 0.9. If the ratio c/v is equal to 0.9, we should look at $p(S = 100\%) = 0.9$. If we want to have zero overflows, then $v = \infty$, and a collection will always occur if the probability is greater than zero. If there is no overflow cost, then $v = 0$, and the bin should only be collected when it is full so as to maximize efficiency. Since we target a compromise solution between collection efficiency and the number of overflows (service level), v should be positive and less than infinity. Different companies/contracts might have different compromises between efficiency and service level. We test two different levels of confidence, representing two different ways of establishing a compromise between efficiency and the number of overflows: 87.5%, which corresponds to a collection/overflow cost ratio of 0.875; and 95%, which corresponds to a collection/overflow cost ratio of 0.95.

3.3. Integration of the predictive methods into the dynamic reverse IRP

The use of data-driven predictive methodologies allows the development of a dynamic reverse IRP capable of defining routing plans that properly account for future events. To achieve this, first, we use one of the two predictive methods to calculate the expected daily filling rate for each bin i on each day $t \in T_{training}$ ($\hat{a}_{it}, \forall\, i \in I \backslash \{n+1\}, t \in T_{training}$); then, we use those rates to feed the dynamic reverse IRP as explained in Fig. 5. The HMM allows to predict the probability of overflowing bins,

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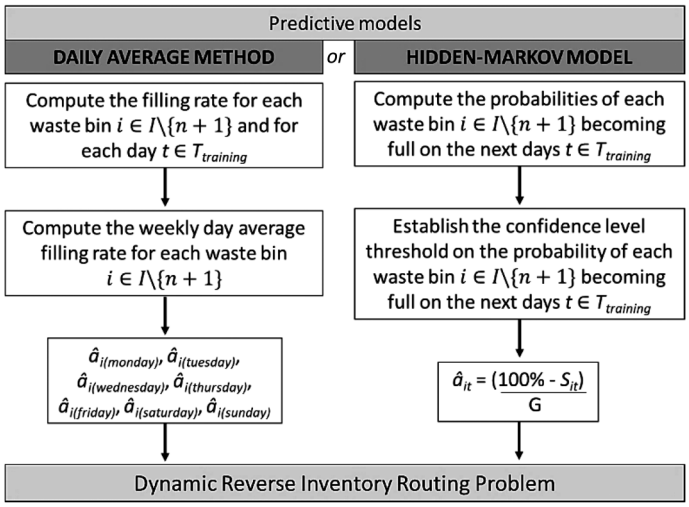

Fig. 5. Integration of predictive methodologies in the dynamic reverse IRP.

and by establishing a confidence level, we are able to estimate different filling rates depending on the sensor information about the fill levels of the waste bins; the daily average method allows to predict different filling rates depending on the day of the week.

In the daily average method, a daily filling rate is calculated for each waste bin $i \in I\backslash\{n+1\}$ and for each day $t \in T_{training}$. An average filling rate for each bin and each day is then calculated. This is assumed to be the expected daily filling rate for each day of the week (i.e., $\hat{a}_{i(monday)}$, $\hat{a}_{i(tuesday)}$, …). The estimates do not change over the testing set, that is, for example, for every Monday of the testing set, the expected filling rate for bin i is the same as calculated for the training set.

In the HMM, the probability of each waste bin becoming full in the next few days, given its current state, is learned from the training set. Therefore, in the testing set, depending on how much waste there is inside a bin (according to the transmitted sensor information, S_{it}), different expected daily filling rates are estimated. The expected daily filling rate of a bin i on day t ($\hat{a}_{it}$) is estimated from the probability matrices computed for observations equal to 0%, 25%, 50%, 75%, and 100%. The fill levels of the waste bins transmitted by the sensors are rounded to the nearest observation level. A threshold parameter is used to establish how confident we are in terms of the number of days needed for the waste bin to become completely full (state 100%). Different confidence levels about the probability of a waste bin becoming full can be considered to calculate the expected daily filling rate. The estimated number of days for a waste bin to become full, given its current state, is denominated by G and varies according to the defined confidence level threshold: the lower the confidence level, the smaller G is. In more conservative scenarios, the expected daily filling rate obtained is higher. Conversely, the higher the confidence level, the less conservative the scenario is, and a lower expected daily filling rate is obtained since the waste bin needs more days to become completely full (larger G). Having the probability matrix for an observed state of waste bin i on day t and the defined confidence level threshold, the expected daily filling rate for that day is the amount of waste that is still necessary for that waste bin to become full, divided by the number of days until this happens (G). Therefore, the expected filling rate for that day is given by

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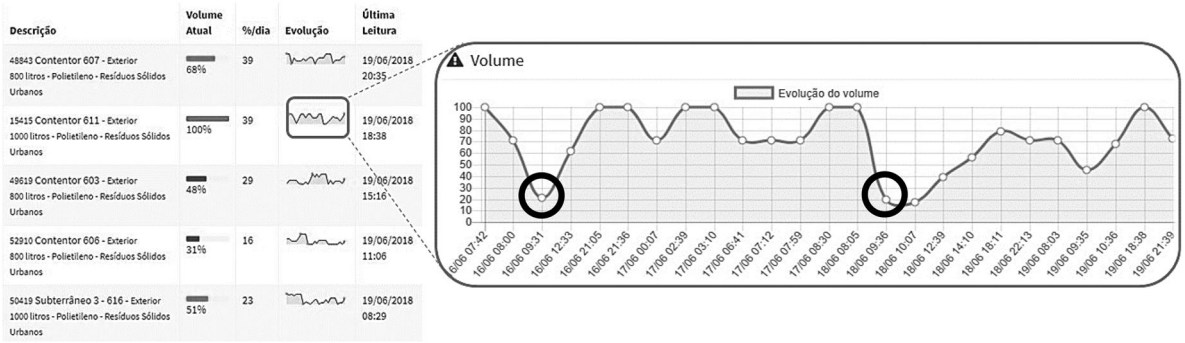

Fig. 6. Sensor information about the fill level of waste bin 611 during four days (*Source*: 360Waste Platform, EVOX, 2018).

$\hat{a}_{it} = (100\% - S_{it})/G$. Each day, the same process is repeated to estimate the expected filling rate for that day.

4. Real-case study with sensorized bins

The proposed D-DRIRP approach is tested using real data provided by a Portuguese waste collection company responsible for collecting undifferentiated waste in the municipality of Castelo Branco. The company is developing a pilot test in which 18 waste bins are equipped with volumetric sensors to transmit real-time information about their fill levels. Sensor data are available since September 2017, and during this period, the company still maintained its pre-defined static routes, performed every Monday, Thursday, and Saturday, collecting all 18 containers regardless of their fill levels.

The company wants to increase the number of sensorized bins and use the available information to improve the efficiency of collection operations and reduce the number of overflows. The processing of the available sensor data and its incorporation into a D-DRIRP approach is explored in our paper to achieve that goal.

4.1. Sensor data

Data about the amount of waste in each waste bin (a volumetric measure, in percentage of the waste bin capacity) is transmitted by the sensors whenever a significant variation in fill level occurs. This generates several transmissions per day for each bin. As such, a large amount of real-time data is available. To illustrate this, Fig. 6 shows the sensor information transmitted from waste bin 611 for four days (from 16 June 2018 to 19 June 2018), where each white dot in the graph corresponds to a transmission. During the four days, only two collections were made—on 16 June, Saturday, and 18 June, Monday—which are highlighted in Fig. 6. Note that there are more reductions in volume, but these are related to the volume of waste being "compacted" when some "heavy" waste is deposited in the bin. These are examples of noisy observations. In this paper, we consider data available from

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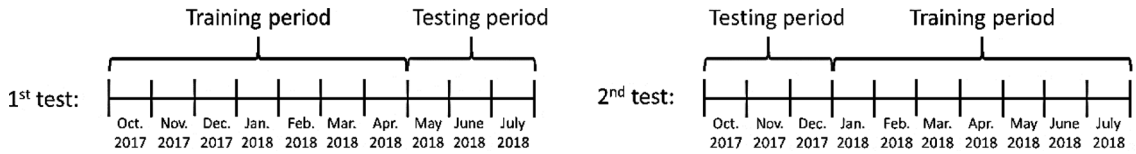

Fig. 7. Training and testing periods considered for the application of the D-DRIRP approach.

October 2017 until the end of July 2018 (10 months; we exclude the first month of the pilot test because some waste bins do not have complete readings for the whole month). As a result, we have 23,522 volumetric sensor measurements—an average of more than 1300 observations per waste bin—to be used by the predictive models. These real data are provided in the online Appendix.

4.2. Simulation

To evaluate our approach, we need to test it in a realistic scenario. As there is not yet the possibility of testing our approach in the real world, we use the real historical data to predict what would have happened if our approach had been used.

In this study, some assumptions are made. Since there is access to sensor information about the waste bin fill levels, we consider the difference between two consecutive sensor observations to calculate the waste deposition rates. If there is a collection in the historical data, but the solution obtained from our approach would not consider a route, the fill level of the waste bin continues to increase. Likewise, if in our solution, we have a collection that is not observed in the historical data, we consider the fill level of the waste bin to be 0. Although this is a simplified approach, we consider it adequate, given that tests are performed considering learning in a set of months (training period) and testing in another set (testing period) (Fig. 7).

To allow a proper validation of the proposed methodology, the 10-month period for which sensor information is available is divided into a training period (seven months) and a testing period (three months). The testing period is then divided into seven-day periods (i.e., weeks, therefore, $T = 7$ days), generating 12 different instances. As can be observed in Fig. 7, two different tests are considered: the first test, where the training period is from 1 October 2017 to 30 April 2018 and the testing period is from 7 May 2018 to 29 July 2018; and the second test, where the training period is from 1 January 2018 to 31 July 2018 and the testing period is from 2 October 2017 to 24 December 2017. These two tests result in a total of 24 different instances. The instances data are also available in the online Appendix.

4.3. Sets and parameters of the mathematical model of the reverse IRP

The sets and parameters values are shown in Table 2.

A set with 18 waste bins and a depot is considered, given by the municipality of Castelo Branco. As the accumulation of undifferentiated waste happens very fast and leads to harmful environmental conditions, the interval between routes is usually small. Thus, for the planning horizon T,

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Table 2
Parameters of the mathematical model of the reverse IRP

Sets	Description	Value	Source
I	Depot and waste bins	One depot and 18 waste bins	Municipality of Castelo Branco
T	Days in the planning horizon	Seven days	–
M	Moments of the day	1–beginning of the day, 2–end of the day	–
Parameters	**Description**	**Value**	**Source**
Q	Vehicle capacity (in kg)	2000 kg	Municipality of Castelo Branco
B	Waste density (in kg/m^3)	123.2 kg/m^3	Gomes et al. (2008)
d_{ij}	Distance between node i and node j (in km)	Real distances	Google Maps
S_{i1}	Fill level of bin i at the beginning of the first day of the planning horizon (in percentage of volume)	Sensor data	360Waste Platform
$\hat{a}_{it}$	Expected daily filling rate of bin i on day t (in percentage)	Dependent on the predictive model	Learned from sensor data
E_i	Capacity of bin i (in m^3)	800 L, 1000 L (0.8 m^3, 1 m^3)	Municipality of Castelo Branco
ε	Very small nondimensional value	0.001	–
AI	Active interval for the rolling horizon	Four days	–
RI	Relaxed interval for the rolling horizon	Three days	–
C	Total weight collected if all waste bins were collected at full capacity every day within the planning horizon	13,976.5 kg	–
R	Total distance traveled if a route where all waste bins are visited was followed every day within the planning horizon	697.2 km	–

a time horizon of seven days is established, based on the experience of the company. Two different moments of the day are considered, namely, the beginning of the day and the end of the day.

The vehicle capacity, Q, is provided by the company. The undifferentiated waste density (B) is based on the work of Gomes et al. (2008). Distances d_{ij} are determined using Google Maps and are the road traveling distances between nodes i and j. The amount of waste in bin i on the first day of the planning horizon, S_{i1}, is the value transmitted by the sensors on Day 1 at 8:00 a.m. for each instance. The expected daily filling rate, $\hat{a}_{it}$, is calculated by the two different predictive models used as explained in Section 3.2. The bin capacity, E_i, is provided by the company and is equal to 800 l (0.8 m^3) for nine containers and equal to 1000 L (1 m^3) for the remaining ones. A very small nondimensional value (ε) equal to 0.001 is used. Regarding the active interval (AI) and relaxed interval (RI) parameters, shorter active intervals lead to a more myopic dynamic reverse IRP approach as discussed in Morais et al. (2019). This is because the approach cannot look far enough ahead. In the aforementioned work, the authors conclude that increasing the actual time interval between routes by two days provides an adequate value for AI. In our case, the average time interval between routes is 2.33 days (as routes are performed on Monday, Thursday, and Saturday); as such, the active interval is set to four days ($AI = 4$ days). This AI value was large enough to

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establish a balance between the adequate future planning period and the computational time of the solution. Since the planning horizon is seven days ($T = 7$ days), the relaxed interval is three days ($RI = 3$ days). Finally, the total weight collected if all waste bins are collected at full capacity every day within the planning horizon, parameter C, is equal to 13,976.5 kg (1996.6 kg collected every day, for seven days), and the total distance traveled if all waste bins are visited every day within the planning horizon, parameter R, is equal to 697.2 km (99.6 km traveled every day, for seven days).

4.4. Scenarios and evaluation metrics

Two different data-driven predictive models are embedded in the D-DRIRP approach, establishing two scenarios to be studied: Scenario 1, with the daily average model embedded in the D-DRIRP, and Scenario 2, with the HMM embedded in the D-DRIRP. For the second scenario, two confidence levels are considered: (a) 95%, representing a less conservative perspective, where the ratio of collection cost to overflow cost is 0.95, and (b) 87.5%, representing a more conservative perspective, where the ratio of collection cost to overflow cost is 0.875. Additionally, the current situation is used for comparison purposes. Summarizing, the following scenarios are evaluated:

- Current situation
- Scenario 1: D-DRIRP approach + daily average predictive model;
- Scenario 2a: D-DRIRP approach + HMM (confidence level = 95%); and
- Scenario 2b: D-DRIRP approach + HMM (confidence level = 87.5%).

These scenarios are studied in the two tests described above (see Fig. 7).

Two key performance indicators are used to assess the performance of the scenarios: efficiency—measured by a kg/ratio, which represents the objective function of the mathematical model, that is, maximizing the waste collected while minimizing the distance traveled, and service level—measured by the number of overflowing bins (waste volume greater than or equal to 100%), which represents a constraint of the mathematical model. The mathematical model avoids expected overflows (constraints 15 and 16), but when the estimates do not match the actual filling rates for a given day, overflows may happen. As these two evaluation metrics are conflicting (improving the service level implies a reduction in the efficiency, as waste bins must be collected more frequently, i.e., while at lower capacity), a compromise solution is desirable, characterized by a high kg per km ratio and a small number of overflowing bins.

As the maximum fill level reported by the sensors was 100%, it is impossible to determine, in the current situation, which bins are at 100% capacity (i.e., $S_i = 100\%$) and which are already past this limit (i.e., $S_i > 100\%$). Because of this, to compute the number of overflowing waste bins for Scenarios 1 and 2, we assume that bins with fill levels greater than or equal to 100% of their total volume are overflowing bins ($S_i \geq 100\%$).

As extra evaluation metrics, and based on the set of operational indicators used by the company, we have total and average values for the:

- amount of waste collected;
- distance traveled;

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Table 3
Tested instances and computational times for optimization

Scenario	# Tested instances	Instance size (# bins)	Computational times (s)		
			Average	Min	Max
Scenario 1	84	18	241	1	3099
Scenario 2a	84	18	98	1	2880
Scenario 2b	84	18	155	1	3160

- number of collected waste bins;
- number of visited empty bins;
- number of full bins (those with fill levels greater than or equal to 87.5% of the bin volume but less than 100%, i.e., $87.5\% \leq S_i < 100\%$).

4.5. Results

The HMM was implemented in the Python programming language, while the mathematical model was implemented using version 24.6.1 of the general algebraic modeling system and solved with version 12.6.3 of the IBM ILOG CPLEX Optimization Studio (CPLEX) on an Intel Xeon CPU X5680 @ 3.33 GHz, considering a limit of 3 600 seconds (1 hour) as the maximum computational time allowed. We point out that training was only used for the predictive methods, and the computational time to run the HMM is negligible (milliseconds). The computational burden of the proposed approach comes from the optimization and therefore we only report the computational time of it.

Two tests were performed, each one with 12 instances. For each instance, the optimization model was solved for seven days, that is, seven times. For detailed results of each instance and each day, we refer to the online Appendix. Table 3 shows the number of tested instances, the size of the evaluated instances, and the average/min/max computational times for each scenario. All tested instances were solved optimally, and no gaps were obtained.

Figure 8 shows the total results for each test (summing the values for all instances) regarding the amount of waste collected, distance traveled, number of waste bins collected, number of empty waste bins visited, and number of full bins. From Fig. 8, it is possible to see that more waste is collected per instance when applying the D-DRIRP approach than when compared to the real situation, even if fewer waste bins are visited. This means that, when applying the D-DRIRP approach, waste bins are visited when they are at a higher capacity. The average number of full waste bins per instance (total values for each test divided by the number of instances) demonstrates that, while in the current situation there is no full waste bin, in Scenarios 1 and 2 an, average of 10 waste bins per instance are full. Another important gain obtained when using the proposed approach is the reduction in the number of empty waste bins visited. While in the current situation about 76% (41 out of 54 waste bins) of the visited waste bins were empty in the first test, and about 32% (23 out of 72 waste bins) of the visited waste bins were empty in the second test, in Scenarios 1 and 2, no empty waste bins are visited.

Figure 9 shows the two key performance indicators for all scenarios and compares them with the current situation, highlighting the obtained gains. Figure 9 presents the total values for the

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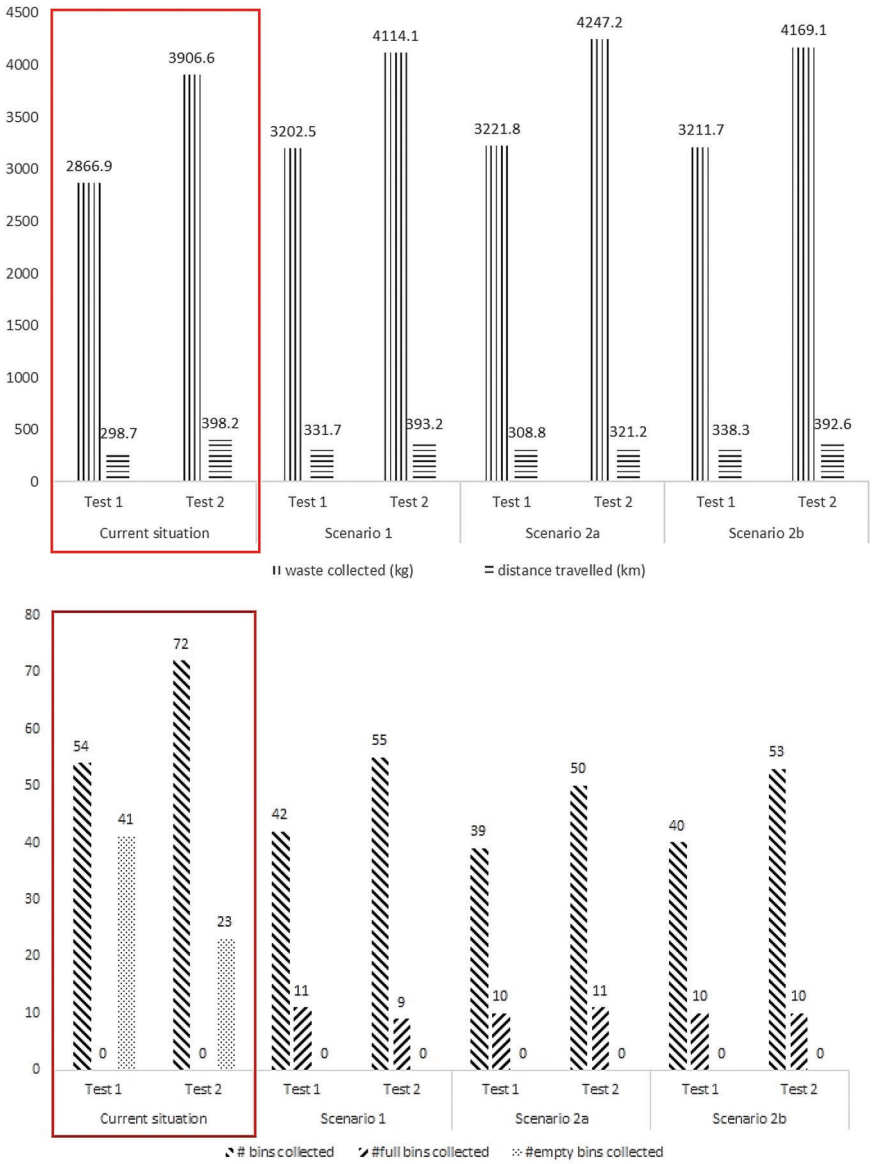

Fig. 8. Results regarding amount of waste collected, distance traveled, number of bins collected, number of full bins, and number of empty bins collected.

two tests, that is, the sum of the values for all instances. From the analysis of Fig. 9, it can be seen that the proposed approach resulted in improvements in all scenarios when compared to the current situation. Namely, the dynamic routing plans are more efficient, and there is a considerable reduction in the number of overflowing bins. These results lead us to conclude that, in the current situation, where static routes are performed, the cost of overflowing is small when compared to the cost of the collection since a significant number of overflowing bins is observed.

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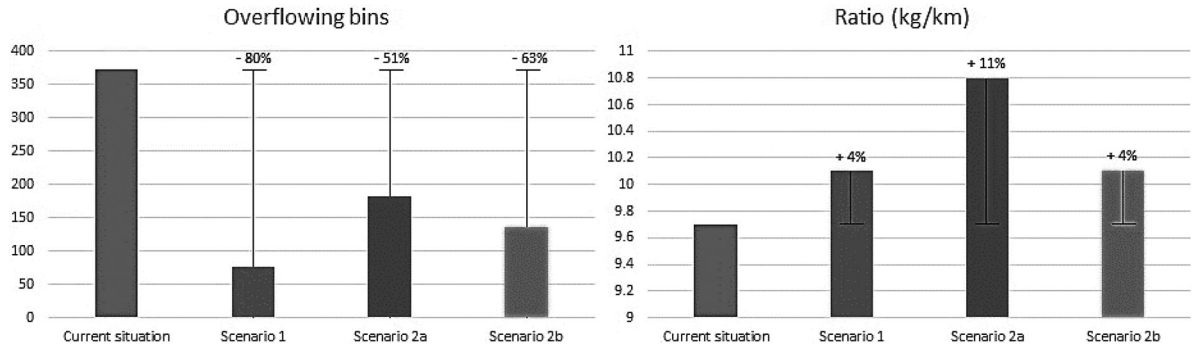

Fig. 9. Comparison between scenarios.

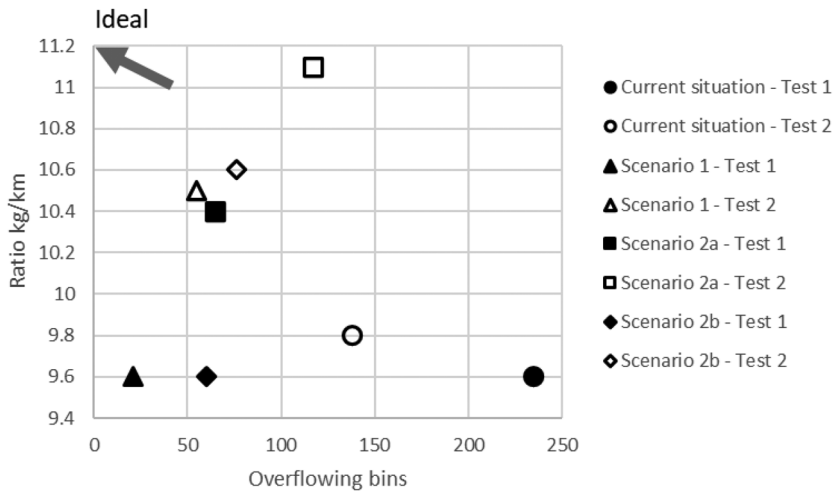

Fig. 10. Summary of results showing the total number of overflowing bins and the kg/km ratio for each test and for each scenario.

Figure 10 details the total results for each scenario divided by test and shows the total number of overflowing bins and the kg/km ratio for the two different tests. When comparing the current situation with the results obtained from the proposed approach, it can be observed that a compromise solution between efficiency and the number of overflowing waste bins is reached with the application of the D-DRIRP approach, as the aim is to increase the kg/km ratio while reducing the number of overflows. The ideal solution would be one with zero overflowing waste bins and a large kg/km ratio (represented by the arrow in Fig. 10). Considering each test, both predictive methods (Scenarios 1 and 2) reduce the number of overflows without decreasing the efficiency of collection. Moreover, the HMM produces solutions with larger (or equal) kg/km ratios when compared to the daily average method, even though it results in a higher number of overflowing bins. This happens because increasing efficiency implies reducing service levels.

Looking specifically at the HMM, this approach allows an objective way to balance the overflow cost and the collection cost based on the defined confidence level (see constraint 25). Scenario 2b,

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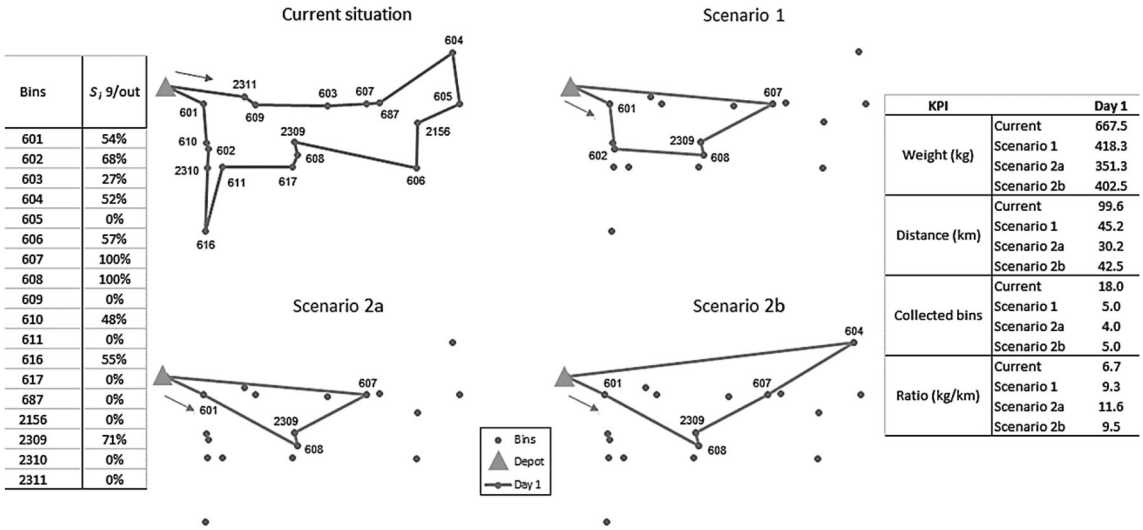

Bins	S_i 9/out
601	54%
602	68%
603	27%
604	52%
605	0%
606	57%
607	100%
608	100%
609	0%
610	48%
611	0%
616	55%
617	0%
687	0%
2156	0%
2309	71%
2310	0%
2311	0%

KPI		Day 1
Weight (kg)	Current	667.5
	Scenario 1	418.3
	Scenario 2a	351.3
	Scenario 2b	402.5
Distance (km)	Current	99.6
	Scenario 1	45.2
	Scenario 2a	30.2
	Scenario 2b	42.5
Collected bins	Current	18.0
	Scenario 1	5.0
	Scenario 2a	4.0
	Scenario 2b	5.0
Ratio (kg/km)	Current	6.7
	Scenario 1	9.3
	Scenario 2a	11.6
	Scenario 2b	9.5

Fig. 11. 9 October routes for all scenarios (illustrative e.g.).

Table 4
Key performance indicators for the second week of October (illustrative e.g.)

	Current situation	Scenario 1	Scenario 2a	Scenario 2b
Routes (#)	3	7	6	7
Weight (kg)	2692.766	2945.84	3124.88	3164.32
Distance (km)	398.24	366.12	307.68	387.94
Collected bins	72	39	31	42
Full bins	0	2	4	3
Overflowing bins	12	3	12	4
Ratio (kg/km)	6.76	8.05	10.20	8.16

wherein a more conservative confidence level was established, presents a higher overflow cost, and therefore results in a smaller number of overflowing bins than Scenario 2a, where a less conservative confidence level was defined.

To illustrate the results, 9 October was selected as an example (from the second test). Figure 11 shows the routes designed for all scenarios for 9 October. The current situation illustrates the collection route that was performed by the company, where eight empty waste bins were visited (bins 605, 609, 611, 617, 687, 2156, 2310, and 2311). Observing the dynamic routes designed using the D-DRIRP approach, no empty waste bins are visited, and consequently the route efficiency is improved. The most efficient route is the one resulting from Scenario 2a (ratio = 11.63 kg/km), where only four waste bins are collected. Compared to Scenario 2a, in Scenario 2b, one more bin is collected; this is because Scenario 2b is more conservative than Scenario 2a, dealing with higher estimated filling rates.

The total set of indicators for the planning period that includes 9 October— the second week of October—are shown in Table 4. In all scenarios studied with the D-DRIRP approach, better results are obtained when compared to the current situation. In Scenarios 1 and 2, a higher amount

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Table 5
Results for computational tests

Instance size (# Bins)	# Tested instances	Computational time (seconds)		
		Average	Min	Max
36	18	376	33	1880
54	18	1454	105	6336
72	18	3482	1151	7200

of waste is collected when compared to the current situation, even if smaller distances are traveled and fewer waste bins are visited, demonstrating that waste bins are collected while at a higher capacity when applying the D-DRIRP approach. Scenario 2 (either 2a or 2b) presents the best ratio; however, in terms of overflowing bins, Scenario 1 has the best performance. In particular, for the second week of October, Scenario 1 presents a smaller number of overflowing bins when compared to Scenario 2b (three bins against four bins) but a smaller ratio (8.05 kg/km vs. 8.16 kg/km). It is worth mentioning that Scenario 2a has the same number of overflowing bins as in the current situation (12 bins) but a much higher kg/km ratio (10.20 kg/km vs. 6.76 kg/km).

From the results obtained, it is possible to state that Scenario 2b represents a good compromise solution between efficiency (high kg/km ratio) and service level (small number of overflowing waste bins). Thus, the HMM predictive method with a more conservative confidence level appears to represent the best compromise among the studied predictive methodologies. Nonetheless, the daily average predictive method also reveals good results, but it does not allow adjusting the level of compromise between efficiency and service level (i.e., the cost of collection vs. the cost of overflow), which is possible in the HMM predictive method (through the establishment of a confidence level).

4.6. Computational tests

To analyze the performance of the proposed methodology, we include additional computational tests to report what happens when larger instances are considered. Using the real data for the 18 bins, we simulate instances with 36, 54, and 72 bins and solve them using the optimization model. Table 5 shows the size of the evaluated instances, the number of tested instances, and the average/min/max computational times.

In this case, a limit of 7200 seconds (two hours) was established as the maximum computational time allowed for the optimization model. Detailed results for computational times and gaps are available in the online Appendix.

The proposed approach was able to solve a common real-world size instance. For instances with 36 bins, the optimization model obtained final solutions in 376 seconds (on average); for instances with 54 bins, this average rises to 1454 seconds. For instances with 72 bins, the model obtained final solutions in an average of 3482 seconds; in this case, for five of 18 instances, the model could not solve to optimality within the maximum time allowed, and gaps were obtained (on average, 7% of gap). These results sustain that the proposed approach can be used in practice by waste management companies since it is able to obtain a route plan in less than 30 minutes (for instances

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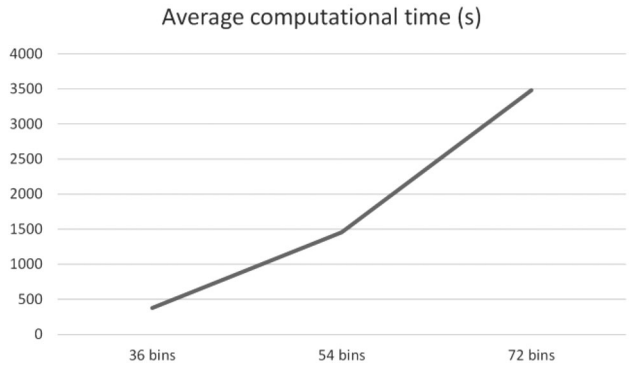

Fig. 12. Average computational time required for the optimization model to solve instances with 36, 54, and 72 bins.

with up to 54 bins). For example, if the sensor information for the 54 bins is received at 8:00 a.m., by 8:30 a.m., a route plan is available for the drivers to execute that day.

Figure 12 illustrates the average computational time required for the model to solve the three different-size instances.

5. Discussion and insights

In this paper, we presented a data-driven approach to solve a dynamic reverse IRP, improving waste collection operations efficiency and service levels. We studied two different predictive methods to estimate waste generation: a daily average method and an HMM model. These methods, when coupled with a dynamic reverse IRP optimization model, allowed for the planning of dynamic routes considering a medium-term planning horizon. The proposed methodology was shown to be efficient and capable of solving large instances, and this research contributed to extend the existing literature on smart waste collection.

By presenting an objective way to balance efficiency and service levels through the definition of the confidence level, the HMM predictive model allowed the definition of a compromise solution. In this way, we can conclude that the use of machine learning techniques in the D-DRIRP approach enabled adjustments in the level of compromise between efficiency and service level. To pursue a "smarter" operation, waste collection companies can take advantage of the large amount of real-time data transmitted by sensors, using efficient techniques such as machine learning to process data and extract patterns and knowledge. When using machine learning techniques to predict overflowing bins and estimate waste generation into the D-DRIRP approach, companies can adjust the desired compromise level between increasing efficiency and reducing the number of overflowing bins by adjusting the collection cost and the overflow cost.

Although the research conducted in this work has proven to be very relevant, continuous updating (or learning) of the models remains important as new data become available. Also, further developments can be still devised, and five main future research lines can be identified. First, about the use of the HMM predictive method, a deep study regarding the confidence level should be performed to identify an "optimal" confidence level capable of making the trade-off between efficiency

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and service level explicit. Second, a possible generalization can be made in the developed approach by associating a cost to each overflowing waste bin, for example, in the form of a fine for violating service level quality. This opens the possibility of exploring a problem that aims to minimize transportation costs and overflowing costs. Third, other types of collection systems should be studied, namely, recyclable waste. The case studied in this paper, undifferentiated waste, is not the best to explore the potential of the developed methodology, as it is characterized by a high daily filling rate and has no associated revenues. In the case of recyclable waste collection, better results could be obtained as it has a market value (and consequently associated revenues) and presents lower filling rates. Finally, an important future line of research is to study which waste bins, from the whole set of bins, should be equipped with sensors, using a cost-benefit analysis.

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